

KUSHAL CHATURVEDI

AI Engineer | Intelligent Systems | Workflow Automation

Hyderabad, Telangana, India

☎ 9982008447 ✉ kushalchaturvedi506@gmail.com 🔗 linkedin.com/in/kushal-chaturvedi

PROFILE SUMMARY

- AI Engineer with **4.5+ years of experience** building **AI-powered automation platforms**, intelligent data systems, and scalable execution frameworks.
- Hands-on experience using **Large Language Models (Claude Sonnet)** for AI-assisted system design, code generation, refactoring, and technical documentation.
- Strong expertise in **Python, Prefect, metadata-driven orchestration**, LLM workflow composition, and parallel execution architectures.
- Proven ability to modernize legacy **COBOL/JCL mainframe workloads** into Python-based intelligent pipelines with measurable performance improvements.

EXPERIENCE

AI Engineer – Intelligent Workflow & Automation Systems

Aug 2023 – Present

Cognizant Technology Solutions

Hyderabad, India

- Designed and developed an **AI-powered workflow orchestration platform** using **Python, Prefect 2.x, and YAML-based metadata**.
- Applied **AI-assisted development** techniques for code generation, architectural reasoning, refactoring, and documentation, reducing development effort by **60%**.
- Engineered a **configuration-driven execution framework** enabling dynamic task creation, dependency resolution, and rule-based validations.
- Implemented **parallel multi-workflow execution** supporting **10+ concurrent workflows** with intelligent retries and failure classification.
- Modernized complex **COBOL/JCL ETL pipelines** by translating business logic into Python-based orchestration workflows with **40% faster execution**.
- Built enterprise-grade data-quality frameworks including schema enforcement, encoding detection (UTF-8/Latin-1), deduplication, and integrity checks processing **500K+ records/day**.

Data Engineer – Mainframe ETL

July 2021 – Aug 2023

Cognizant Technology Solutions

Hyderabad, India

- Developed and maintained large-scale batch ETL pipelines on IBM z/OS using **COBOL, JCL, VSAM, DB2, and SYNCSORT**.
- Implemented delta processing, multi-file joins, balancing, and reconciliation logic for enterprise reporting systems.
- Collaborated with modernization initiatives to migrate legacy logic into cloud-ready Python architectures.

FEATURED AI PROJECTS

AI-Powered Intelligent Workflow Orchestration Platform

AI Engineer

- Built an intelligent orchestration system integrating **Claude Sonnet LLM** to assist with workflow design, validation, and optimization.
- Implemented YAML-driven dynamic task generation enabling zero-code workflow onboarding for non-technical users.
- Achieved **99.5% production uptime** using intelligent retries, execution analytics, and observability.

High-Performance Data Reconciliation Engine

AI Engineer / Developer

- Developed a scalable reconciliation engine using **Python, DuckDB, Pandas, and Parquet** to compare datasets with **5M+ rows**.
- Optimized SQL-driven execution achieving **15x performance improvement**, reducing reconciliation time from hours to minutes.

TECHNICAL SKILLS

AI & LLMs: Claude Sonnet, Prompt Engineering, AI-Assisted Development

LLM Frameworks: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Templates, Tool Calling

Workflow Orchestration: Prefect 2.x, Dynamic Workflow Generation, Parallel Execution

Programming: Python (asyncio, multiprocessing, Pandas, DuckDB, SQLAlchemy, Streamlit)

Data Engineering: PySpark, Spark SQL, ETL/ELT, Parquet, CSV, JSON, Fixed-width

Databases & Storage: PostgreSQL, DuckDB, MySQL, DB2, MongoDB, VSAM

Cloud Platform: Microsoft Azure (ADF, Databricks, ADLS Gen2)

Document Intelligence: PyMuPDF (fitz), PDF Text Extraction, Visual Diffing

Tools: Git, CI/CD Pipelines, VS Code, Jupyter Notebook

EDUCATION

Jamia Millia Islamia

Sep 2017 – May 2021

B.Tech in Mechanical Engineering (9.6 SGPA)

New Delhi, India